Experiment 24

MySQL Queries Implementation for College Admission Form

Aim

To manage student details, courses, admission applications, marks, and admission status using MySQL.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program

Step 1: Create database

```
CREATE DATABASE CollegeAdmissionDB;
USE CollegeAdmissionDB;
```

Step 2: Create tables

```
CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    dob DATE,
    gender VARCHAR(10),
    phone VARCHAR(15),
    email VARCHAR(100)
);

CREATE TABLE Course (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50),
    fees DECIMAL(10,2)
);

CREATE TABLE Admission (
    application_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    marks DECIMAL(5,2),
    admission_status VARCHAR(20),
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);
```

Step 3: Insert sample records

```
INSERT INTO Student VALUES
(1, 'Ravi', '2007-05-12', 'Male', '9876543210', 'ravi@gmail.com'),
(2, 'Priya', '2007-08-20', 'Female', '9876501234', 'priya@gmail.com'),
(3, 'Arun', '2006-12-15', 'Male', '9876512345', 'arun@gmail.com');

INSERT INTO Course VALUES
(101, 'CSE', 75000),
(102, 'AIML', 80000),
(103, 'ECE', 65000);

INSERT INTO Admission VALUES
```

```
(1001, 1, 101, 92.50, 'Approved'),
(1002, 2, 102, 95.00, 'Pending'),
(1003, 3, 103, 78.00, 'Rejected');
```

Queries

Display students

```
SELECT * FROM Student;
```

Display courses

```
SELECT * FROM Course;
```

Search by student name

```
SELECT * FROM Student WHERE student_name = 'Priya';
```

Display admission details

```
SELECT s.student_name, c.course_name, a.marks, a.admission_status
FROM Student s
JOIN Admission a ON s.student_id = a.student_id
JOIN Course c ON a.course_id = c.course_id;
```

Students scoring above 90

```
SELECT * FROM Admission WHERE marks > 90;
```

Update admission status

```
UPDATE Admission SET admission_status = 'Approved' WHERE application_id = 1002;
```

Count approved admissions

```
SELECT COUNT(*) AS approved_admissions FROM Admission WHERE admission_status = 'Approved';
```

Courses with fees below 75000

```
SELECT * FROM Course WHERE fees < 75000;
```

Sort marks descending

```
SELECT s.student_name, a.marks
FROM Student s JOIN Admission a ON s.student_id = a.student_id
ORDER BY a.marks DESC;
```

Delete an application (optional)

```
DELETE FROM Admission WHERE application_id = 1003;
```

Expected Output:

```
1 row in set (0.03 sec)

mysql> CREATE DATABASE CollegeAdmissionDB;
Query OK, 1 row affected (0.04 sec)

mysql> USE CollegeAdmissionDB;
Database changed
mysql> CREATE TABLE Student (
    ->     student_id INT PRIMARY KEY,
    ->     student_name VARCHAR(50),
    ->     dob DATE,
    ->     gender VARCHAR(10),
    ->     phone VARCHAR(15),
    ->     email VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE Course (
    ->     course_id INT PRIMARY KEY,
    ->     course_name VARCHAR(50),
    ->     fees DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

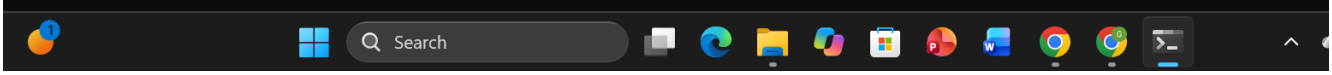
mysql>
mysql> CREATE TABLE Admission (
    ->     application_id INT PRIMARY KEY,
    ->     student_id INT,
    ->     course_id INT,
    ->     marks DECIMAL(5,2),
    ->     admission_status VARCHAR(20),
    ->     FOREIGN KEY (student_id) REFERENCES Student(student_id),
    ->     FOREIGN KEY (course_id) REFERENCES Course(course_id)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Student VALUES
    -> (1, 'Ravi', '2007-05-12', 'Male', '9876543210', 'ravi@gmail.com'),
    -> (2, 'Priya', '2007-08-20', 'Female', '9876501234', 'priya@gmail.com'),
    -> (3, 'Arun', '2006-12-15', 'Male', '9876512345', 'arun@gmail.com');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Course VALUES
    -> (101, 'CSE', 75000),
    -> (102, 'AIML', 80000),
    -> (103, 'ECE', 65000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Admission VALUES
    -> (1001, 1, 101, 92.50, 'Approved'),
    -> (1002, 2, 102, 95.00, 'Pending'),
    -> (1003, 3, 103, 78.00, 'Rejected');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+------------+--------------+------------+--------+------------+-----------------+
| student_id | student_name | dob        | gender | phone      | email           |
+------------+--------------+------------+--------+------------+-----------------+
|          1 | Ravi         | 2007-05-12 | Male   | 9876543210 | ravi@gmail.com  |
|          2 | Priya        | 2007-08-20 | Female | 9876501234 | priya@gmail.com |
```




```
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+------------+--------------+------------+--------+------------+-----------------+
| student_id | student_name | dob        | gender | phone      | email           |
+------------+--------------+------------+--------+------------+-----------------+
|          1 | Ravi         | 2007-05-12 | Male   | 9876543210 | ravi@gmail.com  |
|          2 | Priya        | 2007-08-20 | Female | 9876501234 | priya@gmail.com |
|          3 | Arun         | 2006-12-15 | Male   | 9876512345 | arun@gmail.com  |
+------------+--------------+------------+--------+------------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Student
    -> WHERE student_name = 'Priya';
+------------+--------------+------------+--------+------------+-----------------+
| student_id | student_name | dob        | gender | phone      | email           |
+------------+--------------+------------+--------+------------+-----------------+
|          2 | Priya        | 2007-08-20 | Female | 9876501234 | priya@gmail.com |
+------------+--------------+------------+--------+------------+-----------------+
1 row in set (0.00 sec)

mysql> SELECT s.student_name, c.course_name,
    ->        a.marks, a.admission_status
    -> FROM Student s
    -> JOIN Admission a ON s.student_id = a.student_id
    -> JOIN Course c ON a.course_id = c.course_id;
+--------------+-------------+-------+------------------+
| student_name | course_name | marks | admission_status |
+--------------+-------------+-------+------------------+
| Ravi         | CSE         | 92.50 | Approved         |
| Priya        | AIML        | 95.00 | Pending          |
| Arun         | ECE         | 78.00 | Rejected         |
+--------------+-------------+-------+------------------+
3 rows in set (0.00 sec)

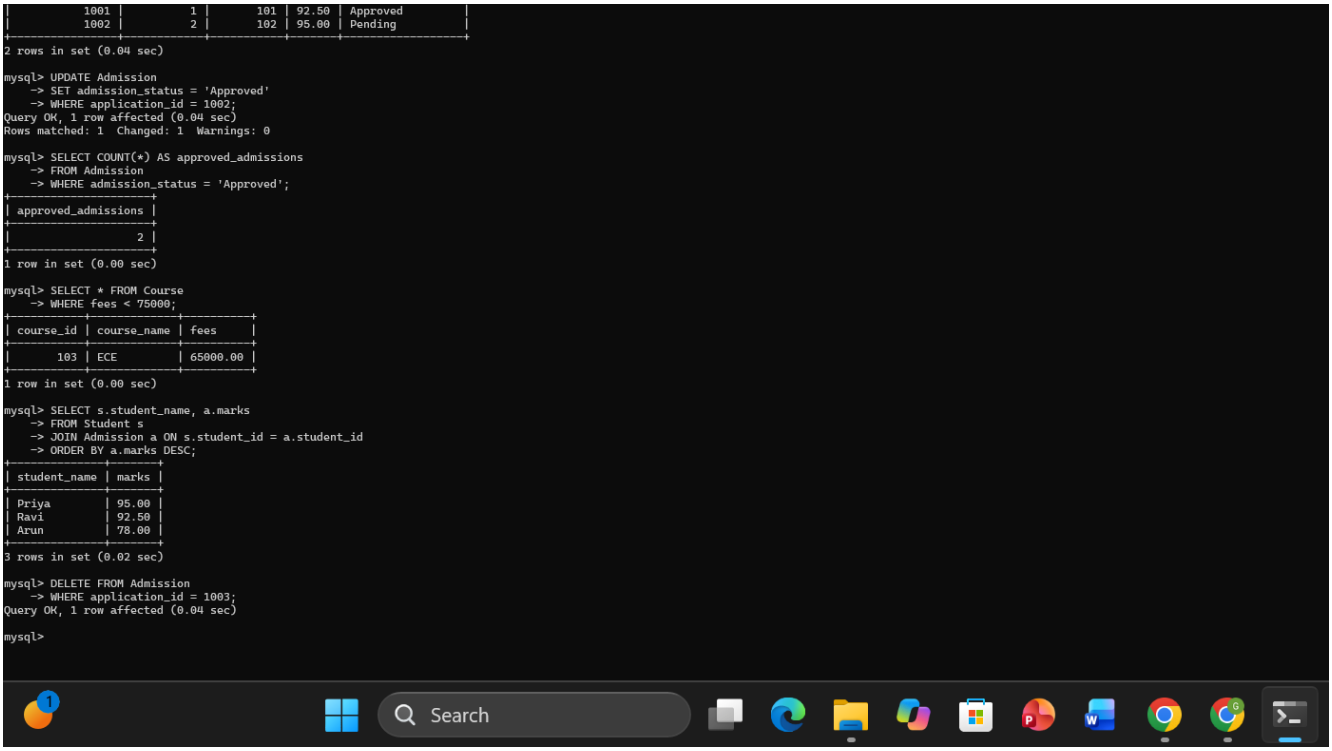
mysql> SELECT * FROM Admission
    -> WHERE marks > 90;
+----------------+------------+-----------+-------+------------------+
| application_id | student_id | course_id | marks | admission_status |
+----------------+------------+-----------+-------+------------------+
|           1001 |          1 |       101 | 92.50 | Approved         |
|           1002 |          2 |       102 | 95.00 | Pending          |
+----------------+------------+-----------+-------+------------------+
2 rows in set (0.04 sec)

mysql> UPDATE Admission
    -> SET admission_status = 'Approved'
    -> WHERE application_id = 1002;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT COUNT(*) AS approved_admissions
    -> FROM Admission
    -> WHERE admission_status = 'Approved';
+----------------------+
| approved_admissions  |
+----------------------+
|                    2 |
+----------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM Course
    -> WHERE fees < 75000;
+-----------+-------------+----------+
| course_id | course_name | fees     |
+-----------+-------------+----------+
```





Observation

The database stores student registration, course details, marks, and application status. Queries search, filter, update, sort, count, and delete records.

Result

Thus, MySQL queries for a college admission form were implemented successfully.

Note

Execute the optional DELETE query only if you want to permanently remove the sample application.